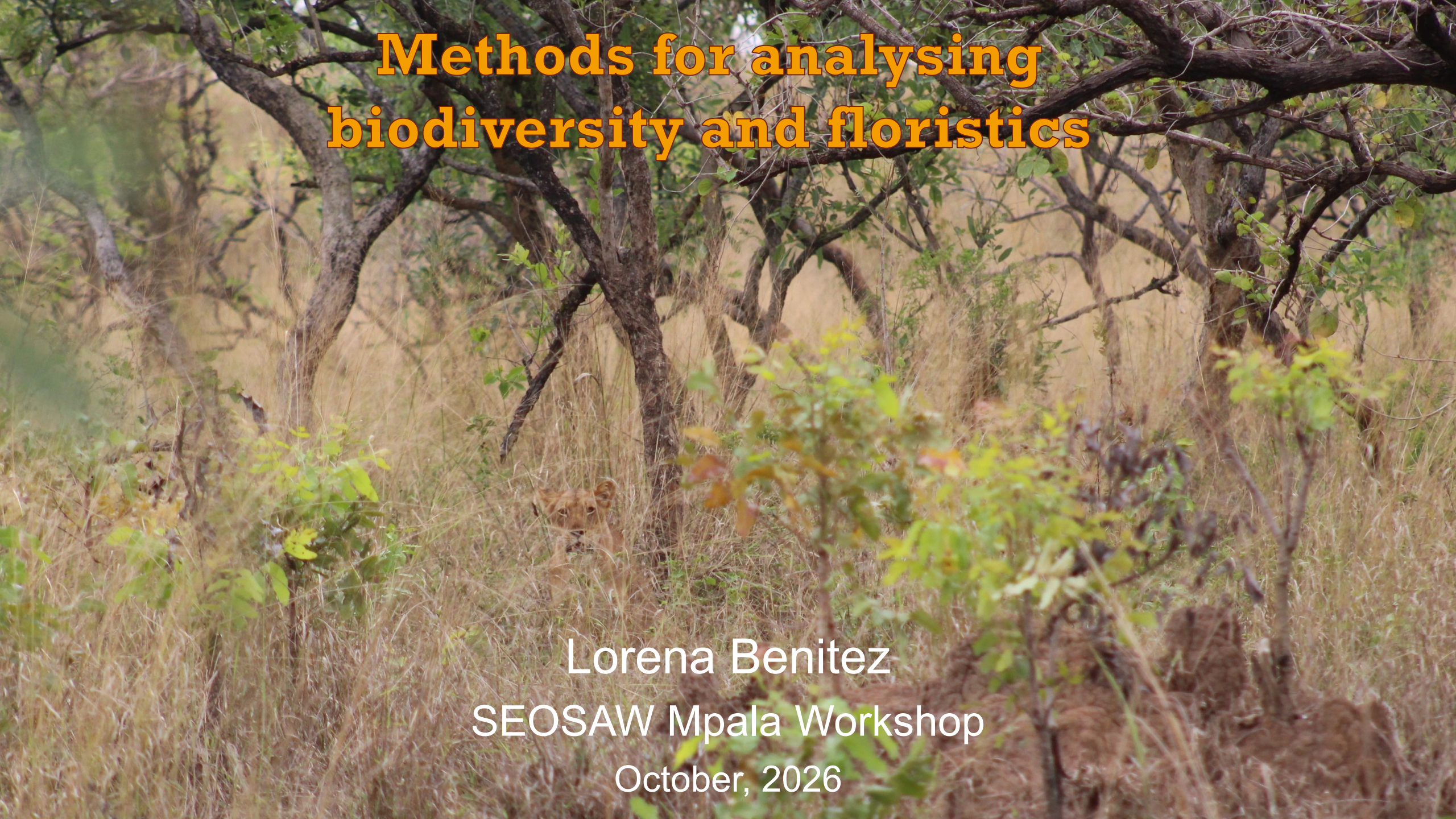


Methods for analysing biodiversity and floristics

Lorena Benitez

SEOSAW Mpala Workshop

October, 2026

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Biodiversity and floristics

- Ecosystems services provided by trees are dependent on species-specific traits
- Thus, knowing the number of species, abundance of each species, and overall community composition (biodiversity) is important for understanding and managing ecosystems
- Floristics is the study of how plant communities and their distributions

Scripts for today

- **01_Calculate_metrics**
  - How many species are there?
- **02_Rarefaction**
  - How does sampling impact diversity?
- **03_Ordination_clustering**
  - How can we define ecological communities?
- **04_Indicators**
  - What species are critical for identifying ecological communities?
  - Do not attempt without first doing script 3

Data for floristics

- Need a lot of plots for floristics
- Thus, we focus on single census plots
- Today we are using ~400 plots from Mozambique



Ecological Diversity

- We want to understand how many different ‘things’ exist in a plot, habitat, land use, or landscape

- Diversity can be estimated on different levels including species, abundances, and function



Diversity metrics

- Species richness
  - How many species are in a plot
- Evenness
  - How are abundances balanced between species



Diversity metrics

- The following metrics all balance richness and evenness

- Shannon diversity
  - Uncertainty of predicting the species of a random individual
  - More sensitive to rare species
- Simpson diversity
  - Probability any two individuals are the same species
  - More sensitive to common species
- Inverse Simpson
  - Probability any two individuals are different species

Diversity metrics

- Rarefied richness
  - Standardises richness by sample size
  - On a plot level, this means the species diversity if every plot had the same number of trees
  - For SEOSAW, we usually rarefy to 25 stems, but for today's practical I chose 10

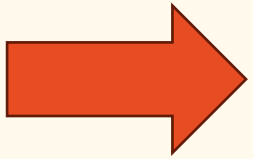
R packages in script 1

- dplyr
  - For data cleaning
- vegan
  - For calculating metrics
- ggplot2
  - For visualising
- Optional: patchwork
  - For making complex plots

Data formatting

- Start with stem data from SEOSAW plots
- For all analyses today, we will need to reformat this into species abundance matrices where rows are plots and species are columns

plot_id	stem_id	species_name_clean	diam_adj
MAR_230	5	Burkea africana	22.09706
MAR_230	20	Indet indet	11.54795
MAR_230	27	Indet indet	19.59706
MAR_230	33	Brachystegia boehmii	24.89706
MAR_230	34	Julbernardia globiflora	16.79706
MAR_230	35	Syzygium guineense	10.94795
MAR_230	42	Brachystegia boehmii	32.80976

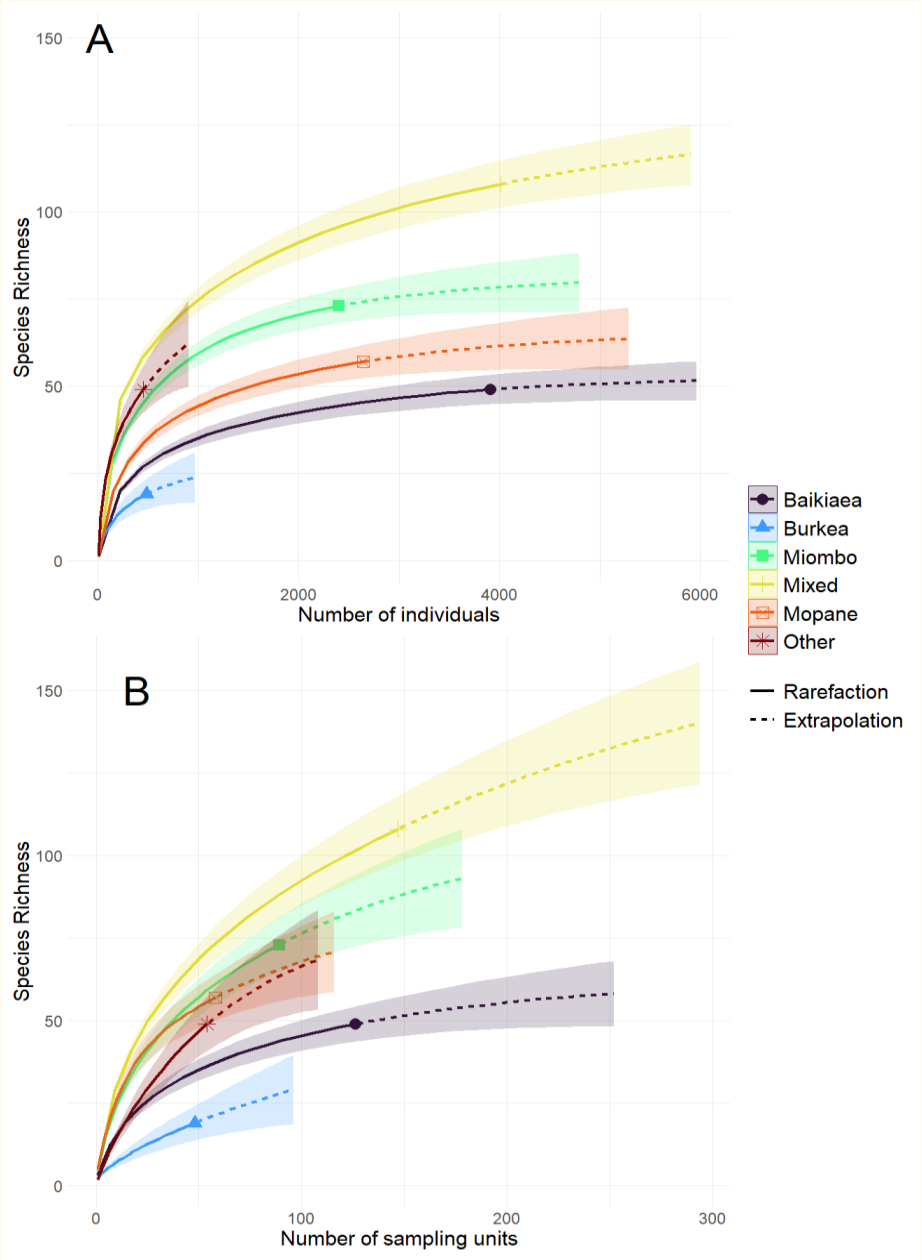


	Albizia amara	Cercidoideae indet	Combretum collinum	Combretum indet
MAR_1	7	8	3	1
MAR_10	0	0	0	0
MAR_100	0	0	1	0
MAR_102	0	0	2	0
MAR_103	0	0	0	0
MAR_104	0	0	0	0
MAR_105	0	0	0	0
MAR_106	0	0	0	0



Rarefaction curves

- Useful to examine how species richness increases with sampling
- Can be used to determine if you have done enough sampling in a specific habitat or treatment



R packages

- Vegan
  - For simple rarefaction
- iNext
  - It can extrapolate to predict the diversity with more plots/stems
  - It allows easy analysis of site level diversity and comparison
  - It allows calculation of rarefaction using multiple metrics
  - It automatically generates confidence intervals
  - It automatically uses ggplot2, so making nice figures is a lot easier



Ordination

- Techniques for reducing complex data into fewer dimensions

- In our case, this means taking data on species abundances and summarising it in fewer dimensions

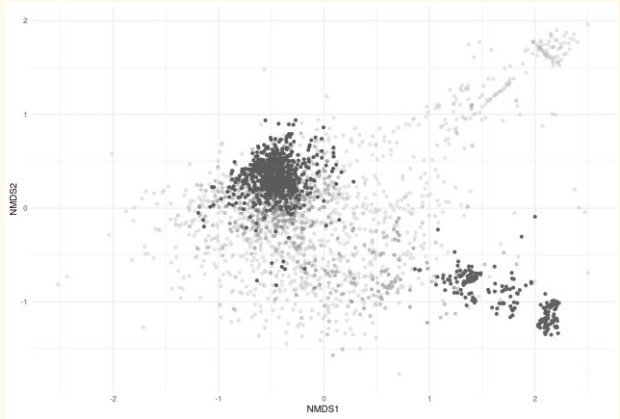
Identifying ecological communities

plot_id	Adansonia	Afrocanthium	Afzelia	Aganope
ABG_1	0	0	0	0
ABG_10	0	0	0	0
ABG_11	0	0	0	0
ABG_12	0	0	0	0
ABG_13	0	0	0	0
ABG_14	0	0	0	0
ABG_15	0	0	0	0
ABG_16	0	0	0	0
ABG_18	0	0	0	0
ABG_19	0	0	0	0
ABG_2	0	0	0	0
ABG_3	0	0	0	0
ABG_4	0	0	0	0
ABG_5	0	0	0	0
ABG_6	0	0	0	0
ABG_7	0	0	0	0
ABG_8	0	0	0	0
ABG_9	0	0	0	0

Generate an abundance matrix

Reduce dimensionality

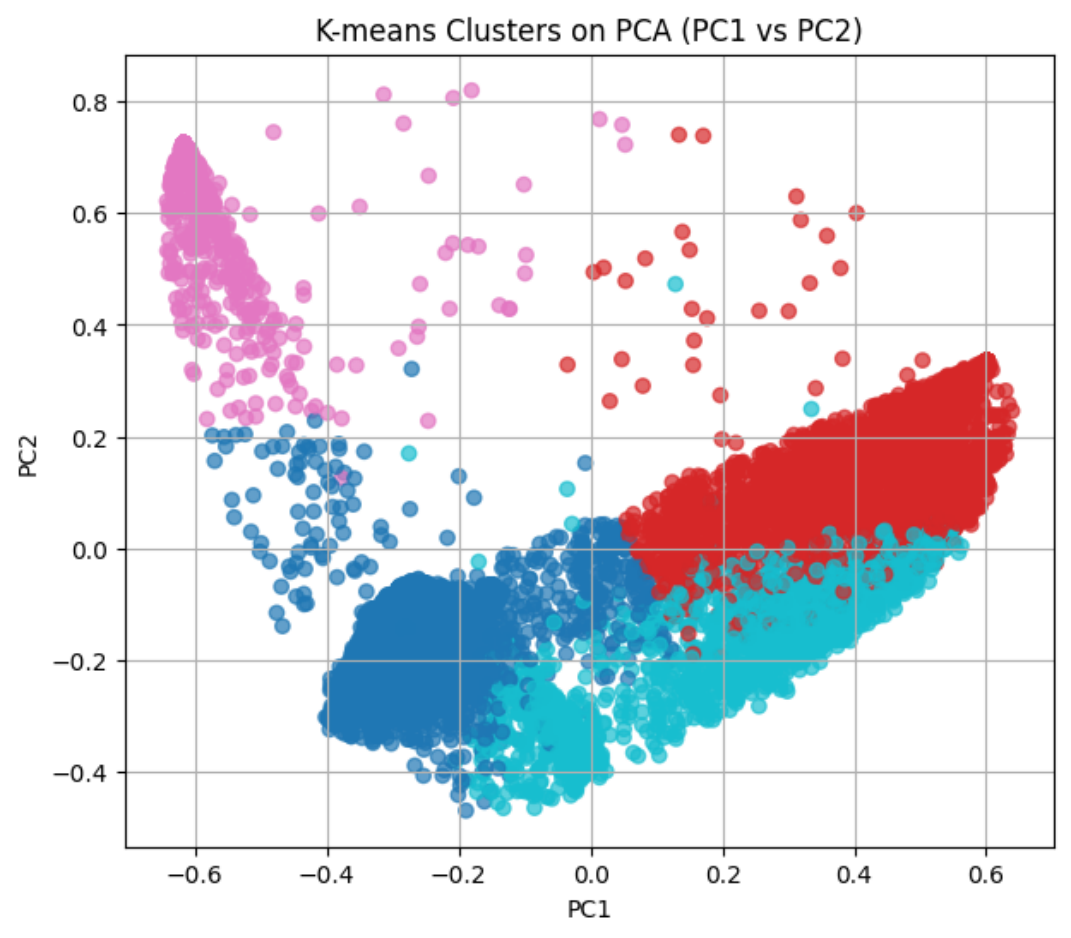
Cluster data





Principal Components Analysis (PCA)

- The classic ordination technique
- Linear
- Still produces many dimensions
- Easily interpretable



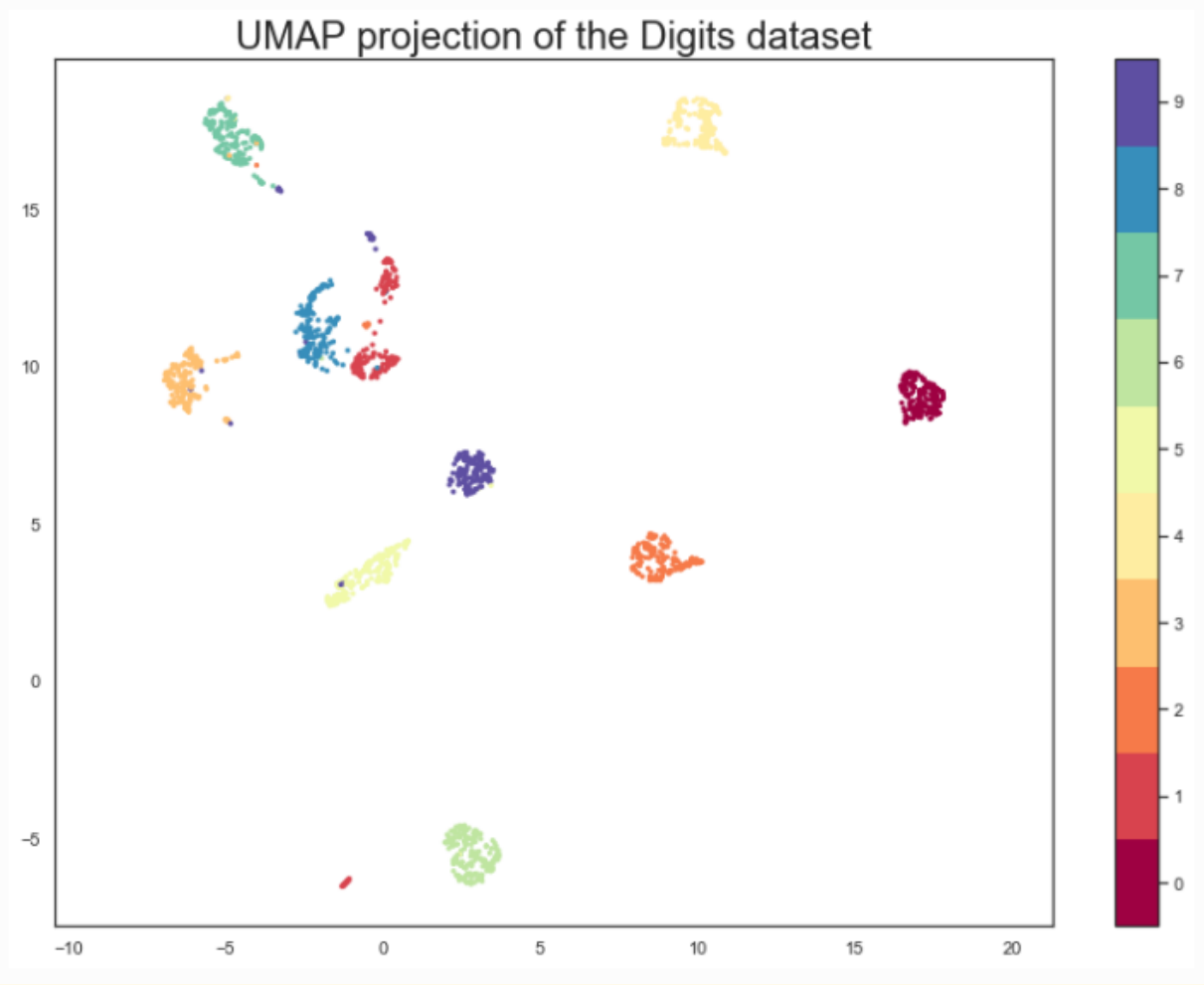
Non-metric Multidimension Scaling (NMDS)

- Non-parametric ordination method
- Based on distance or dissimilarity
- Preserves rank, not values
- The go-to for ecologists
- Very sensitive to outliers



UMAP

- Method for non-linear dimension reduction
- Computationally efficient
- Good documentation
- In my experience, better at defining ecologically meaningful clusters than other methods



https://umap-learn.readthedocs.io/en/latest/index.html

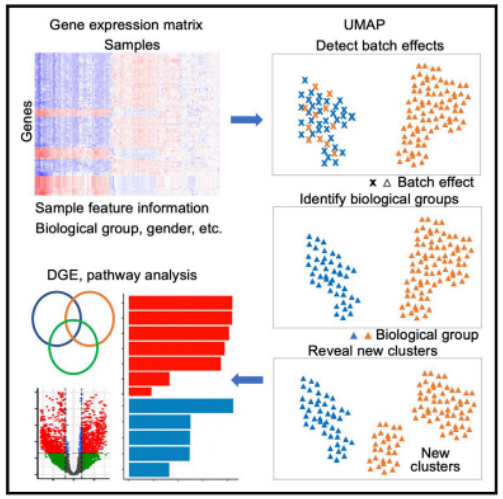
UMAP in biology & ecology

Resource

Cell Reports

Dimensionality reduction by UMAP reinforces sample heterogeneity analysis in bulk transcriptomic data

Graphical abstract



Authors

Yang Yang, Hongjian Sun, Yu Zhang, ..., Yuchen Yang, Di Wu, Di Yu

Correspondence

dwu@unc.edu (D.W.),
di.yu@uq.edu.au (D.Y.)

In brief

Yang et al. compare four major dimensionality reduction methods (PCA, MDS, t-SNE, and UMAP) in analyzing large bulk transcriptomic datasets. UMAP is overall superior to PCA and MDS and shows some advantages over t-SNE in differentiating batch effects, identifying pre-defined biological groups, and revealing in-depth clusters in two-dimensional space.

Highlights

- Four methods, PCA, MDS, t-SNE, and UMAP, are evaluated on 71 bulk transcriptomic datasets
- UMAP is overall superior to PCA and MDS and shows some advantages over t-SNE
- UMAP can efficiently and effectively reveal clusters in two-dimensional space
- Clusters revealed by UMAP are associated with biological features and clinical traits

Ecological Informatics 86 (2025) 103058

Contents lists available at ScienceDirect

Ecological Informatics

journal homepage: www.elsevier.com/locate/ecolinf

Leveraging citizen science to classify and track benthic habitat states: An unsupervised UMAP-HDBSCAN pipeline applied to the global reef life survey dataset

Clément Violet [a,*], Aurélien Boyé [a], Stanislas Dubois [a], Graham J. Edgar [b], Elizabeth S. Oh [b], Rick D. Stuart-Smith [b], Martin P. Marzloff [a]

Ecological Informatics 72 (2022) 101909

Contents lists available at ScienceDirect

Ecological Informatics

journal homepage: www.elsevier.com/locate/ecolinf

Method for passive acoustic monitoring of bird communities using UMAP and a deep neural network

Gabriel Morales [a], Víctor Vargas [b], Diego Espejo [b], Víctor Poblete [b,d,*], Jorge A. Tomasevic [c], Felipe Otondo [d], Juan G. Navedo [e,f]

Science of the Total Environment 815 (2022) 152365

Contents lists available at ScienceDirect

Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

The application of Uniform Manifold Approximation and Projection (UMAP) for unconstrained ordination and classification of biological indicators in aquatic ecology

Djuradj Milošević [a,*], Andrew S. Medeiros [b], Milica Stojković Piperac [a], Dušanka Cvijanović [c], Janne Soininen [d], Aleksandar Milosavljević [e], Bratislav Predić [e]

tSNE

- Focuses on local structure only
- Unstable
- Slow
- Over clusters
- Cannot transform data

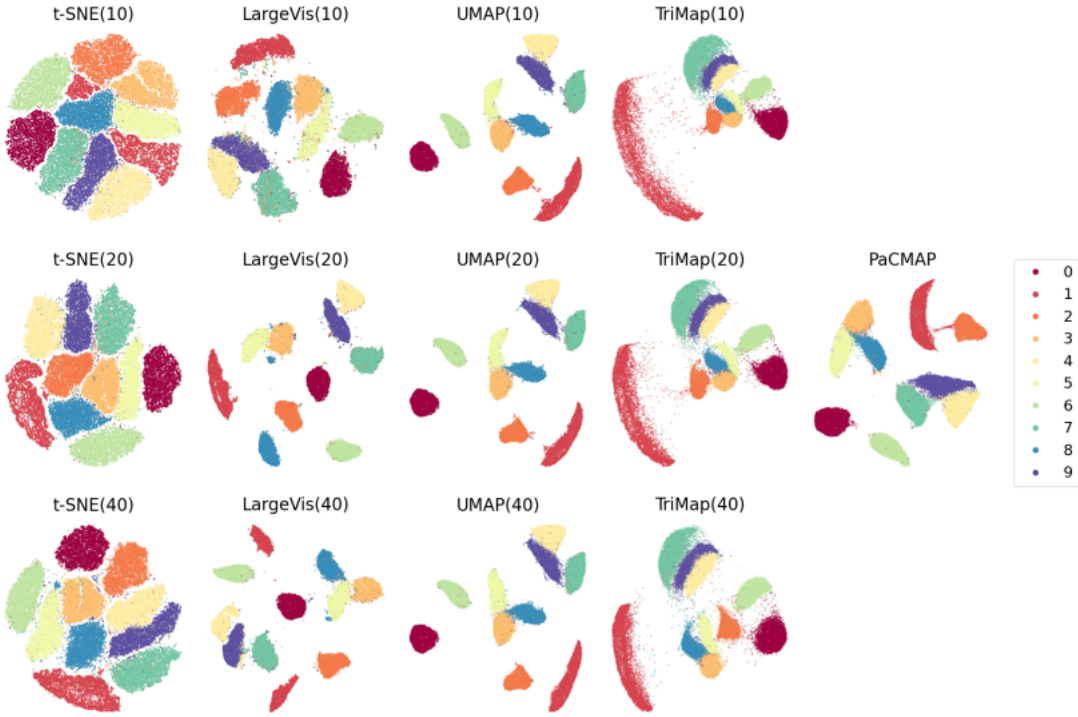
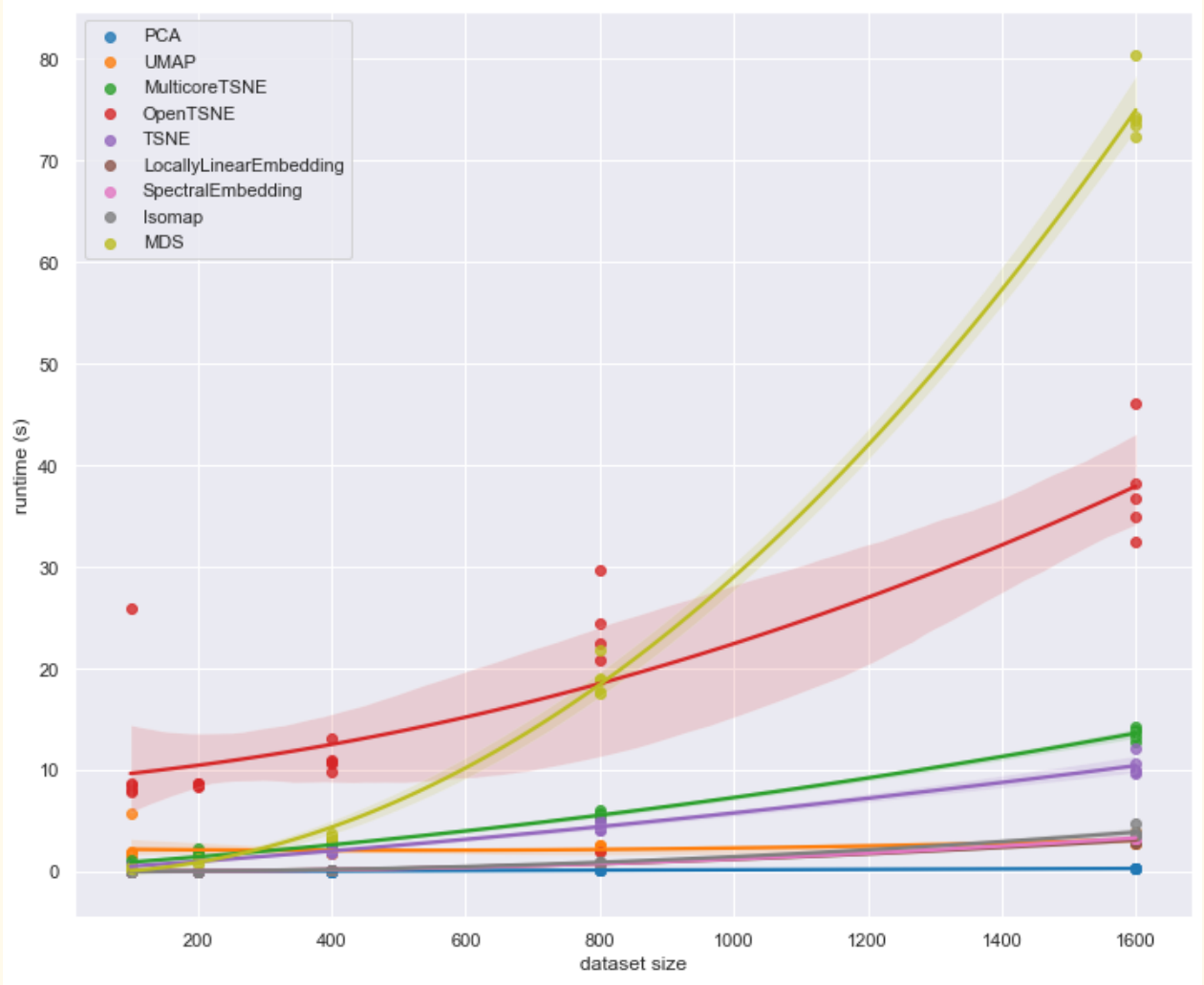


Figure 2: Application of t-SNE, UMAP, TriMap and PaCMAP to the MNIST dataset. Algorithms that preserve local structure tend to separate MNIST classes better. Similar to the experiment in Figure 1, we tuned the most important parameter controlling the visual effect for each algorithm, which is perplexity for t-SNE and LargeVis, n_{NB} for UMAP, and n_{inlier} for TriMap. The values we considered are in the parentheses.

Wang et al. 2021

Efficiency with data



https://umap-learn.readthedocs.io/en/latest/benchmarking.html

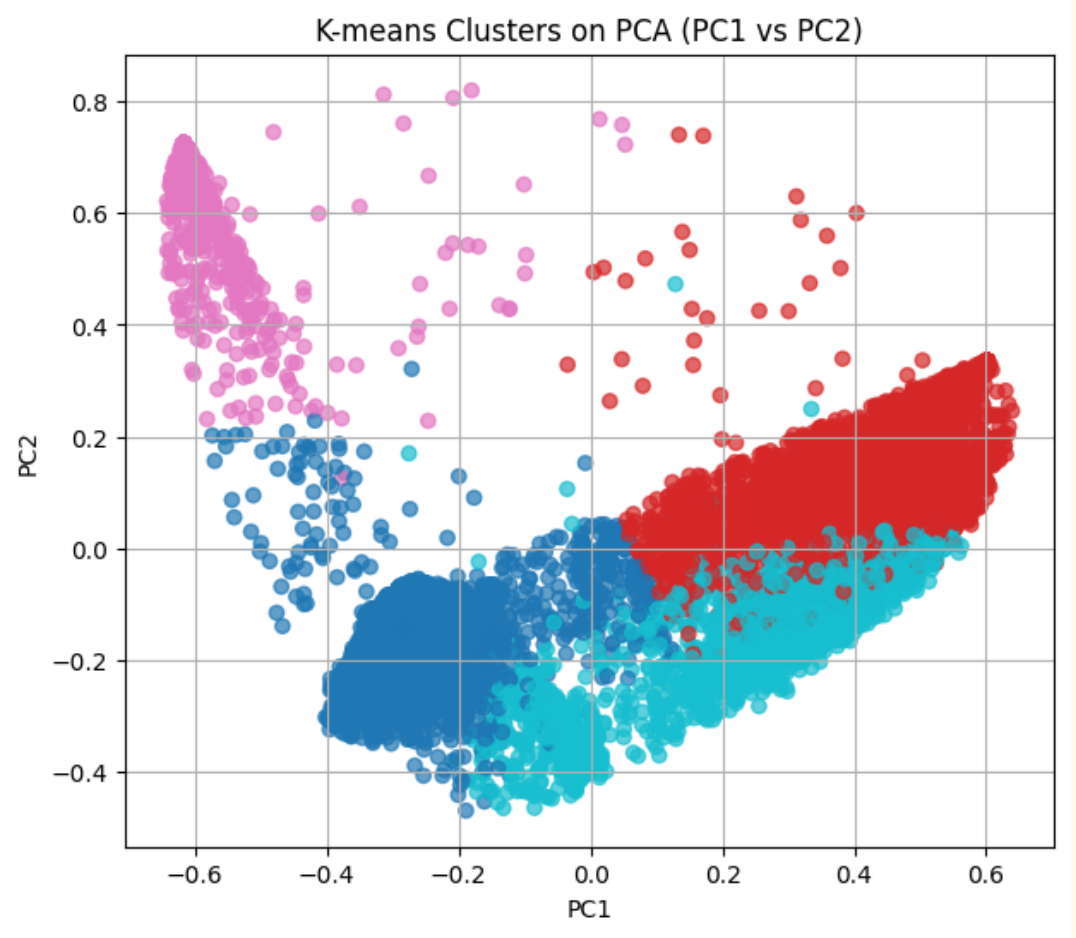
R packages

- *Base R*
  - for PCA
- *factoextra*
  - For visualising PCA
- *vegan*
  - For NMDS
- *umap*
  - For UMAP



K-means clustering

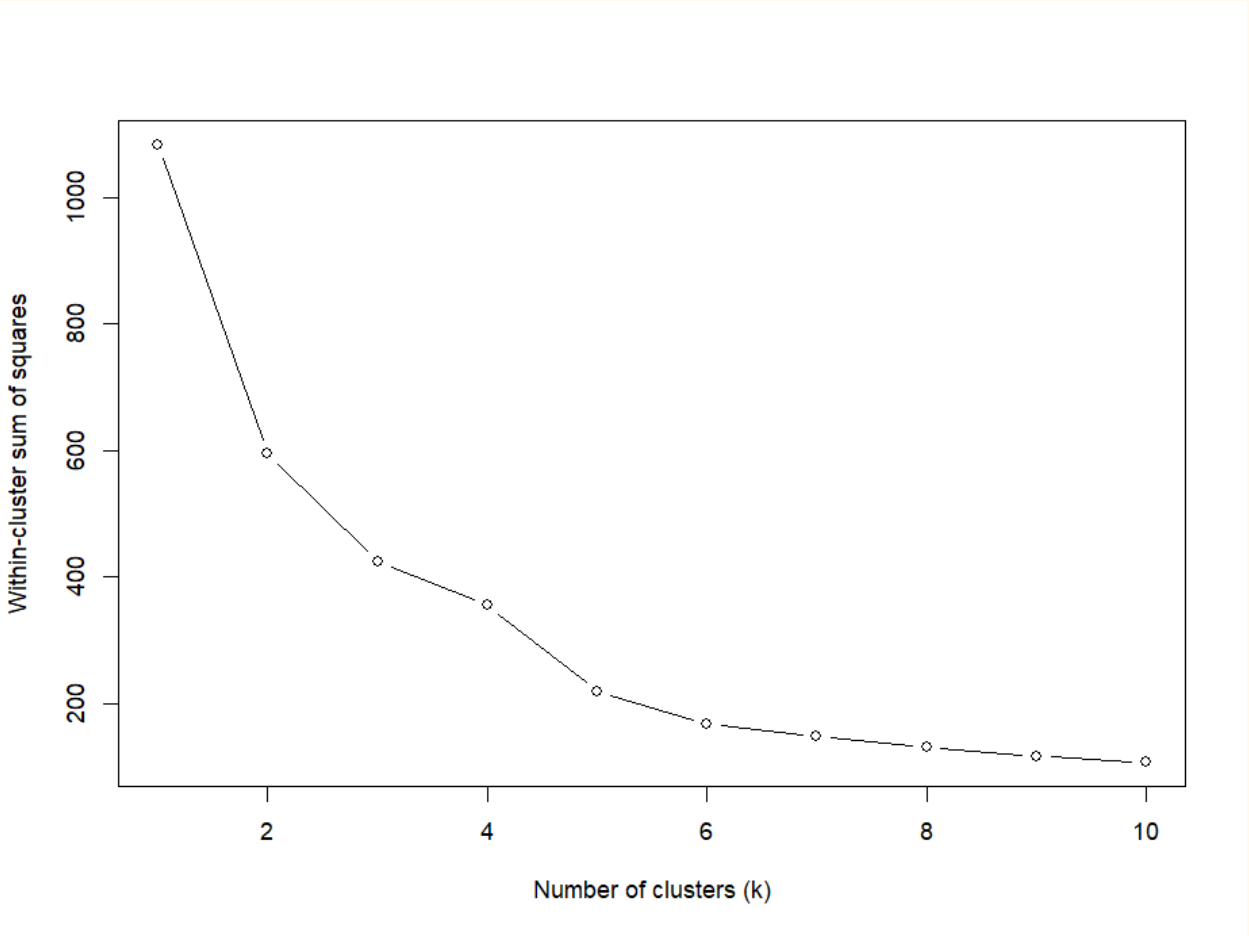
- K = number of clusters
- You choose the K value
- Data points are grouped in K clusters based on location in ordination space



Finding the best K- Elbow method

- Test for best number of clusters
- Use metric called within-cluster sum of squares
- The optimal number of K is where the plot begins to bend (called the elbow)

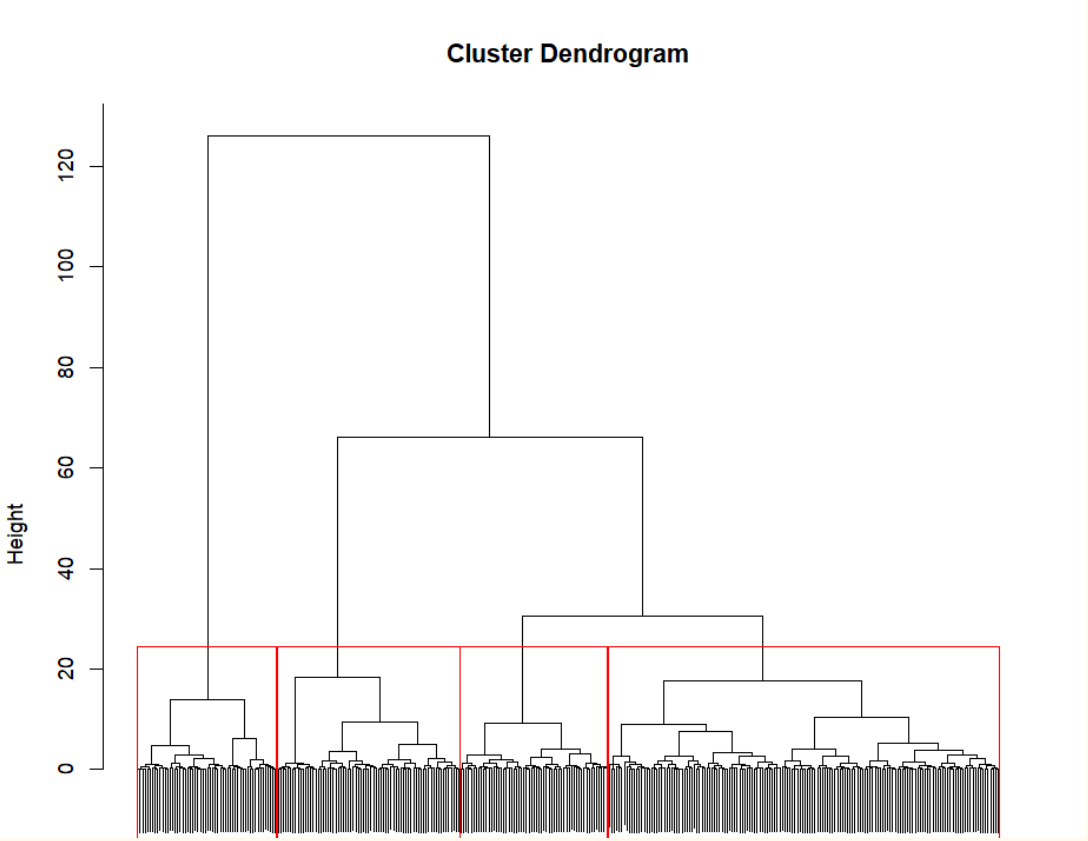
- Always visually check that your K makes sense on the ordination space by plotting the points with colours based on K



Hierarchical clustering

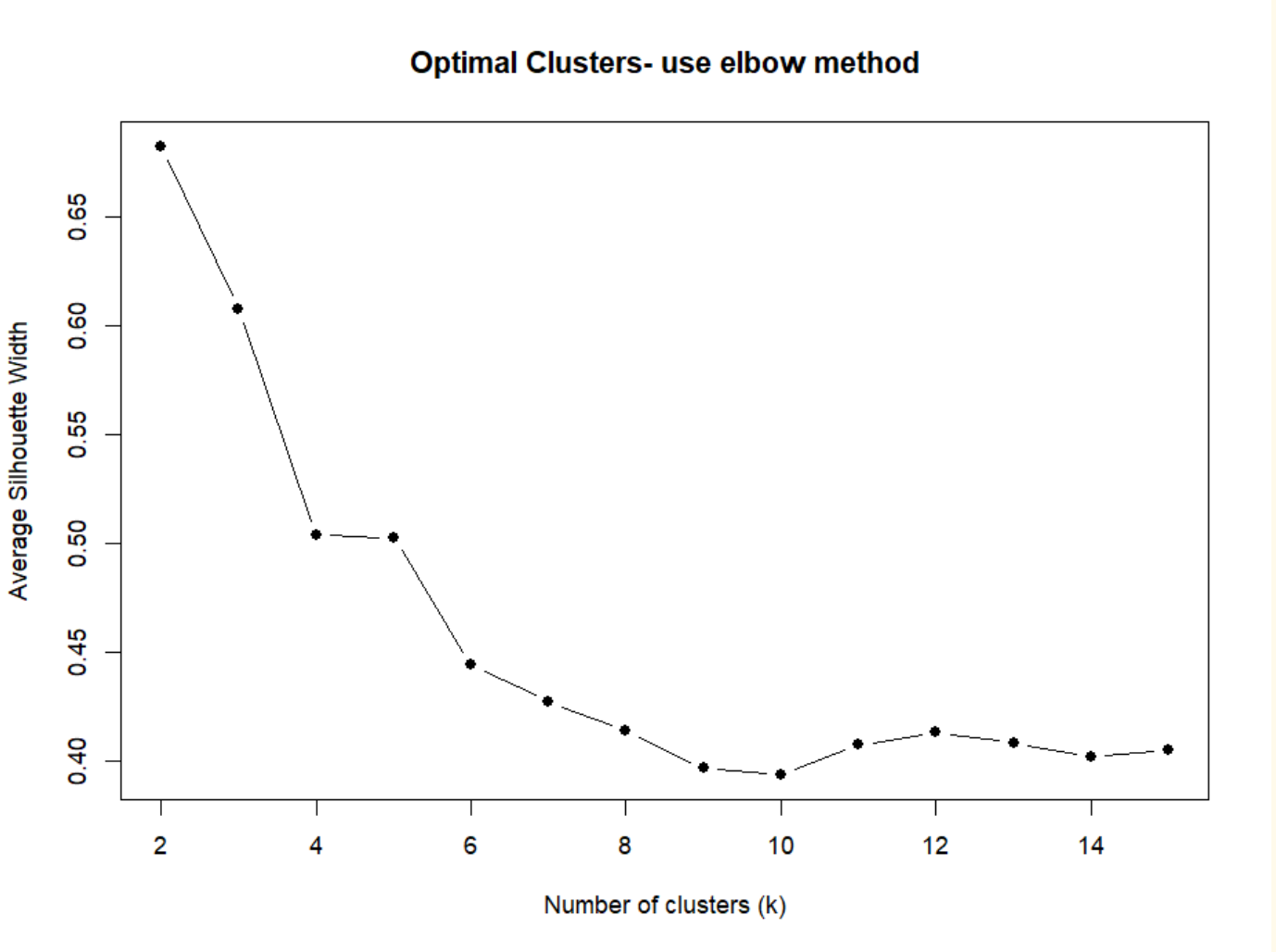
- Creates a tree where plots that are more similar are closer together

- Clusters can easily be subdivided to look at multiple levels of ecological communities



Finding the best K- Silhouette score

- Do not use the elbow method!
- You are looking for the highest average silhouette score
- Higher average silhouette width is, the better defined and separated the clusters are



Problems with ordination and clustering

- You can only identify communities that you have sampled well
- Defining floristic clusters is sensitive to methods and data
- Weakly connected data points (aka outliers) contribute to instability
- This instability is sometimes attributed to ‘noise in the data’, but community composition exists as a continuum where noise is only relevant to any particular solution

Tozer & Keith, 2023



Indicator species

- Clustering only identifies communities

- We need quantitatively define them by determining which species are associated with which communities

indicspecies R package

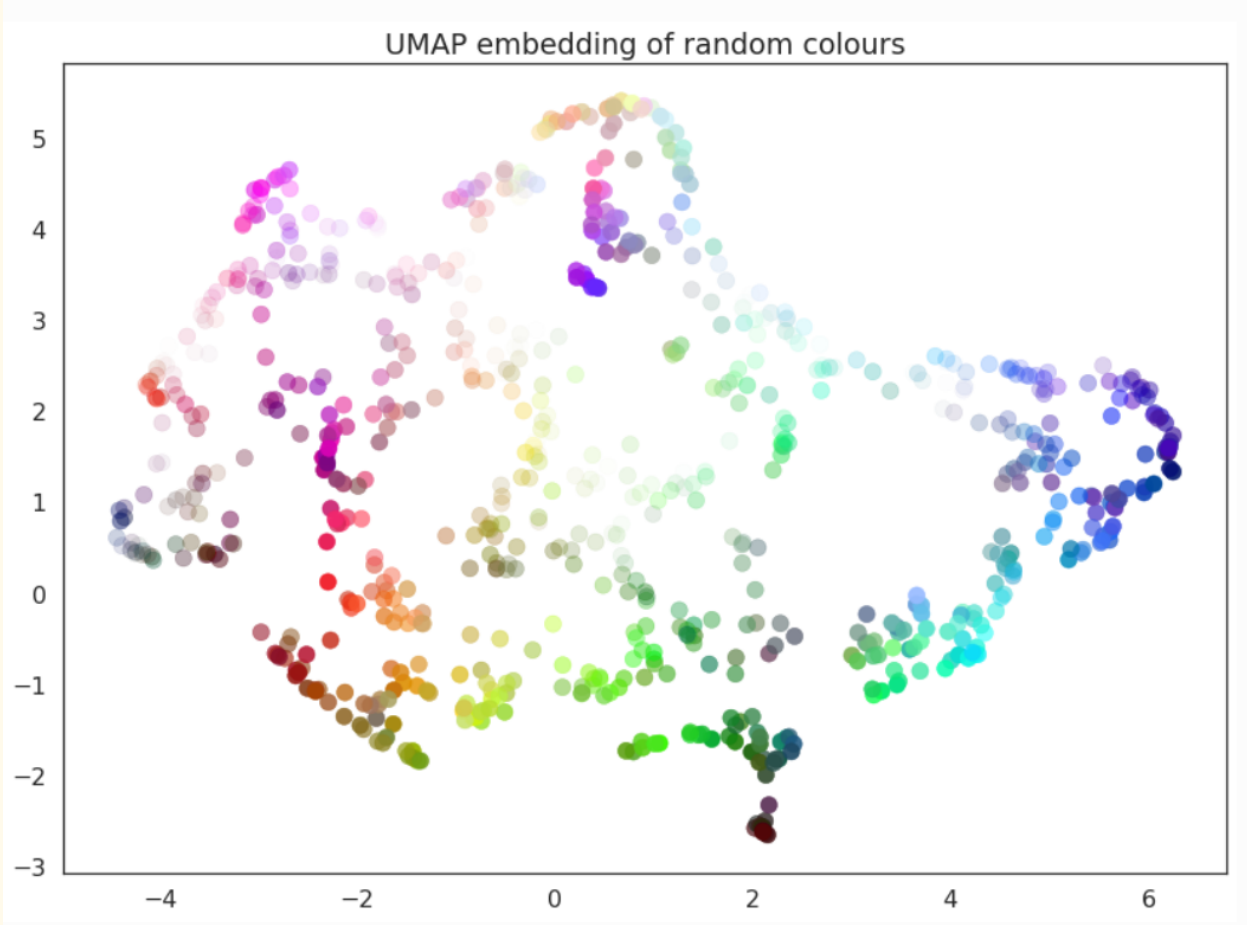
- Inputs: species abundance matrix and cluster assignments
- Combines sensitivity (frequency of species at a site) and specificity (abundance at one cluster compared to others) to determine which genera are strongly associated with each cluster
- Provides an indicator statistic (0-1) to quantify how associated a species is with clusters

Bonus slides



UMAP parameters

- N components
  - Number of dimensions
- Metric
  - Distance method
- Minimum distance
  - (Between points)
- N neighbours
  - Local v. global



Metrics

- Default is Euclidian
- There are lots of options!
- You can also precompute your distances

Minkowski style metrics

- euclidean
- manhattan
- chebyshev
- minkowski

Miscellaneous spatial metrics

- canberra
- braycurtis
- haversine

Normalized spatial metrics

- mahalanobis
- wminkowski
- seuclidean

Angular and correlation metrics

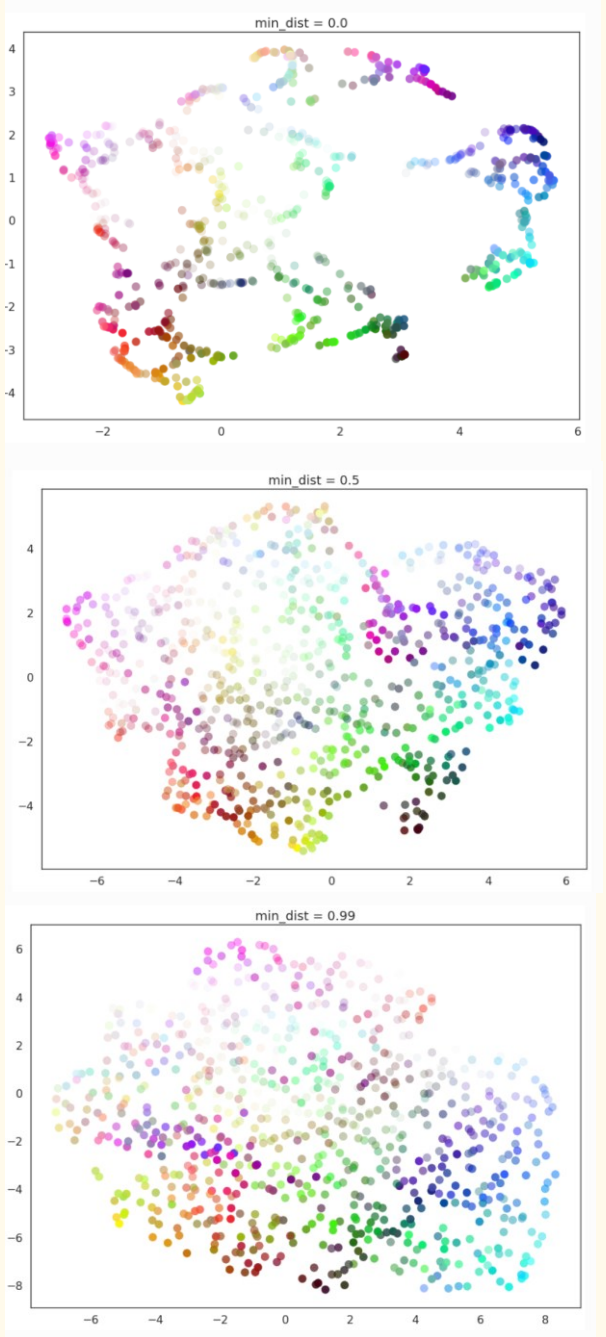
- cosine
- correlation

Metrics for binary data

- hamming
- jaccard
- dice
- russellrao
- kulsinski
- rogerstanimoto
- sokalmichener
- sokalsneath
- yule

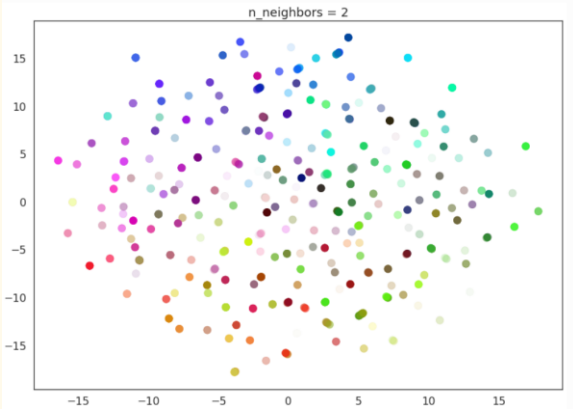
Minimum distance

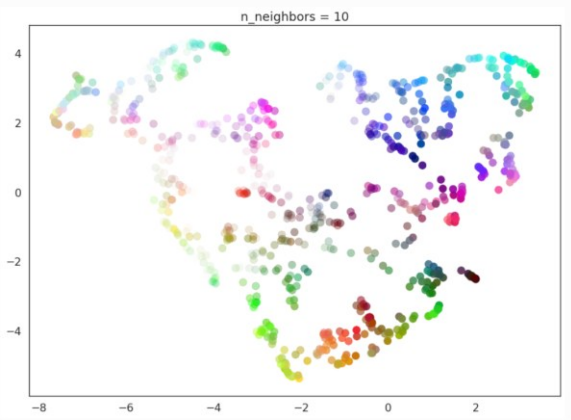
- The minimum distance apart that points are allowed to be in the low dimensional representation
  - Controls how tightly UMAP is allowed to pack points together
- This means that low values = clumpier embeddings
- Larger values preserve of the broad topological structure instead

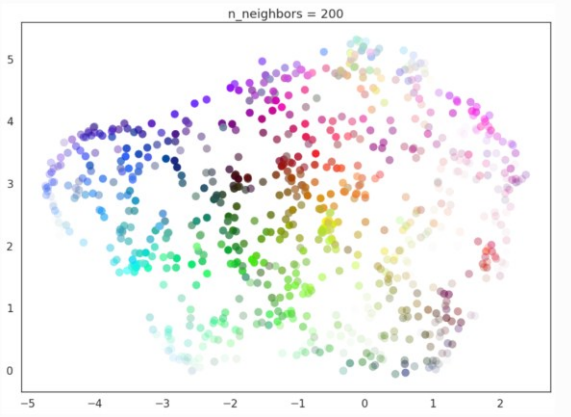


Number of neighbours

- Balances local with global structure
- Low values = more local
- High values = more global



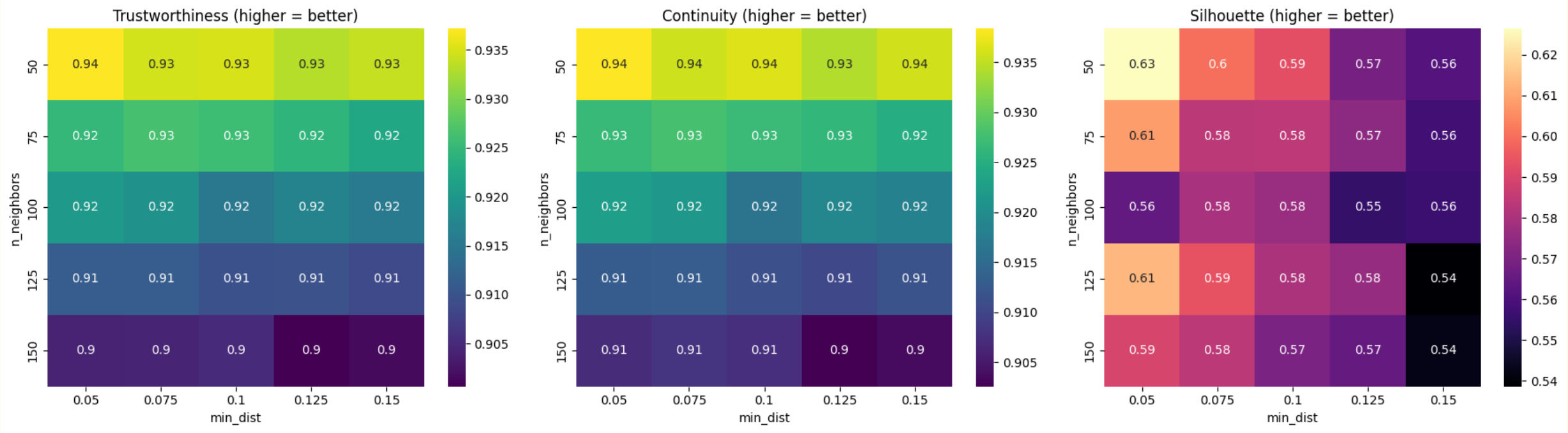




Selecting parameter values

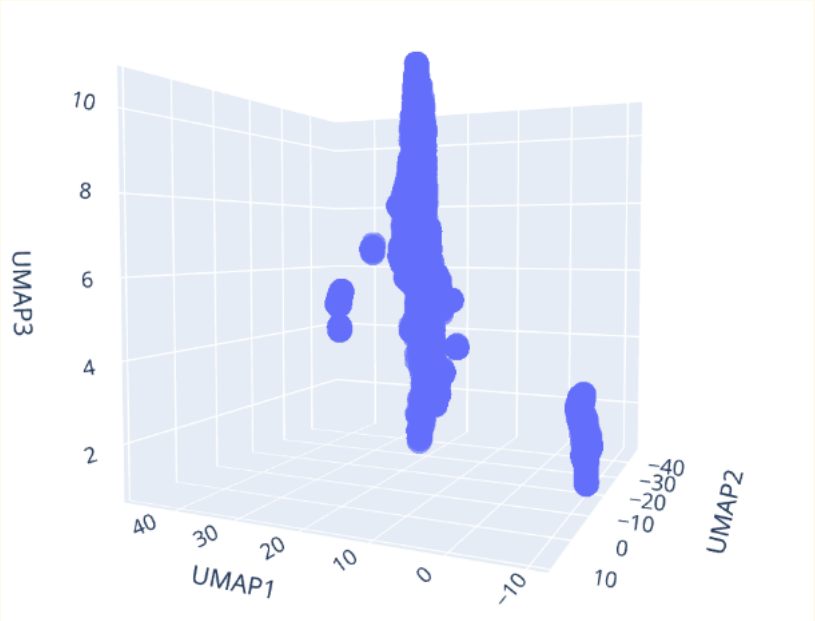
- Visual assessment is best
- UMAP does not provide parameter selection tools
- Trustworthiness
  - How well local neighbourhood is preserved
- Continuity
  - Test if global geometry is reliable
- Silhouette scores
  - Looks at clusters
  - How separated are they?

Parameter selection



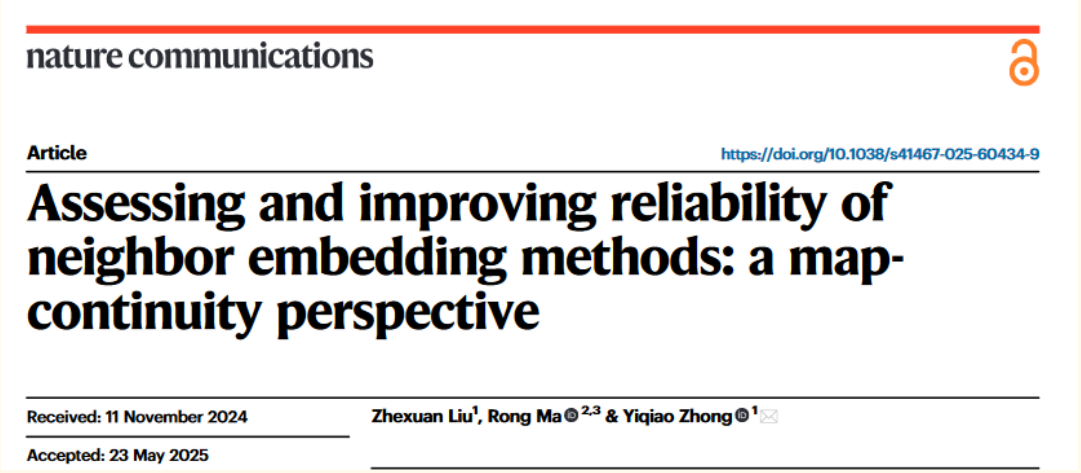
Parameter selection advice

- Approach with caution
- Parameters tend toward preserving local structure instead of global
- Always sense check your parameters by visualising your data



Assessing your UMAP environment

- UMAP and t-SNE can introduce misleading visual artifacts
- This paper suggests metrics for assessing your data for:
  - Perturbation: how uncertain is the classification
  - Singularity: sensitivity of points
  - Perplexity: parameter balancing local and global structure


nature communications

Article

https://doi.org/10.1038/s41467-025-60434-9

Assessing and improving reliability of neighbor embedding methods: a map-continuity perspective

Received: 11 November 2024

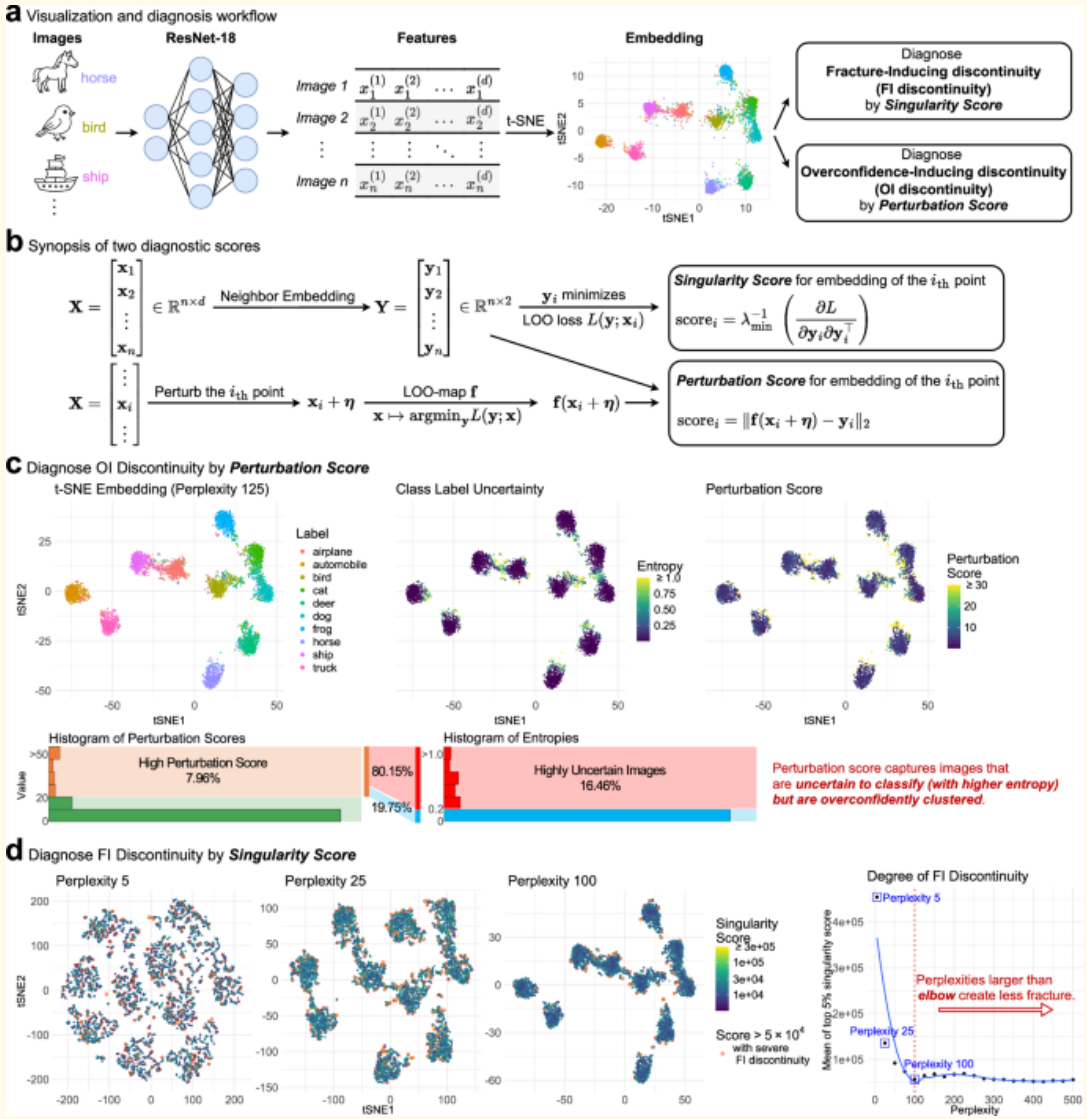
Accepted: 23 May 2025

Zhexuan Liu[1], Rong Ma[2,3] & Yiqiao Zhong[1]

https://www.nature.com/articles/s41467-025-60434-9

Assessing your UMAP environment

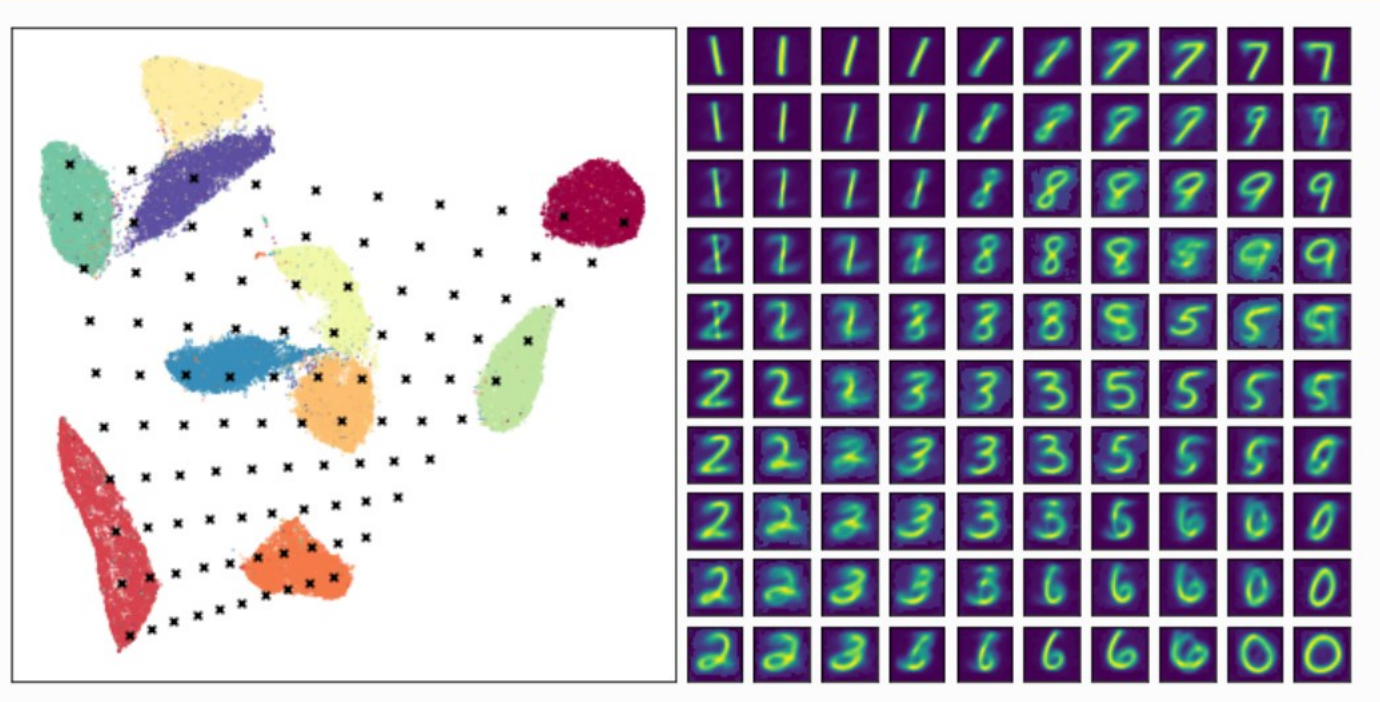
- UMAP and t-SNE can introduce misleading visual artifacts
- This paper suggests metrics for assessing your data for:
  - Perturbation: how uncertain is the classification
  - Singularity:
  - Perplexity: parameter balancing local and global structure



https://github.com/zhexuandliu/MapContinuity-NE-Reliability

Other functions

- Transform
- Inverse transform
- Parametric UMAP (neural network)
- Supervised dimension reduction
- And many more!



Python

- Works great
- Good documentation
- Stick with python if you can

```
# Run UMAP
# Run best UMAP model based on selected parameters
umap_model = UMAP(
    n_neighbors=10,
    min_dist=0.05,
    n_components=3,
    metric="braycurtis",
    verbose=True,
    random_state=17
)

embedding = umap_model.fit_transform(df_remove)
```

```
/usr/local/lib/python3.12/dist-packages/umap/umap_.py:1952: UserWarning:

n_jobs value 1 overridden to 1 by setting random_state. Use no seed for parallelism.

UMAP(min_dist=0.05, n_components=3, n_jobs=1, n_neighbors=10, random_state=17, verbose=True)
Fri Feb 20 14:20:17 2026 Construct fuzzy simplicial set
Fri Feb 20 14:20:17 2026 Finding Nearest Neighbors
Fri Feb 20 14:20:17 2026 Building RP forest with 9 trees
Fri Feb 20 14:20:17 2026 NN descent for 12 iterations
     1  /  12
     2  /  12
     3  /  12
     4  /  12
     5  /  12
    Stopping threshold met -- exiting after 5 iterations
Fri Feb 20 14:20:18 2026 Finished Nearest Neighbor Search
Fri Feb 20 14:20:18 2026 Construct embedding
Epochs completed: 100%| 500/500 [00:33]
    completed  0  /  500 epochs
    completed  50  /  500 epochs
    completed  100  /  500 epochs
    completed  150  /  500 epochs
    completed  200  /  500 epochs
    completed  250  /  500 epochs
    completed  300  /  500 epochs
    completed  350  /  500 epochs
    completed  400  /  500 epochs
    completed  450  /  500 epochs
Fri Feb 20 14:20:52 2026 Finished embedding
```

R

- uwot
  - Works well, but has limitations
  - Need additional package for some metrics
  - Limited ability to transform data
  - Cannot handle precomputed distance matrices
- umap
  - Wrapper for python package
  - Does not contain all of the functions

Summary of UMAP

Benefits

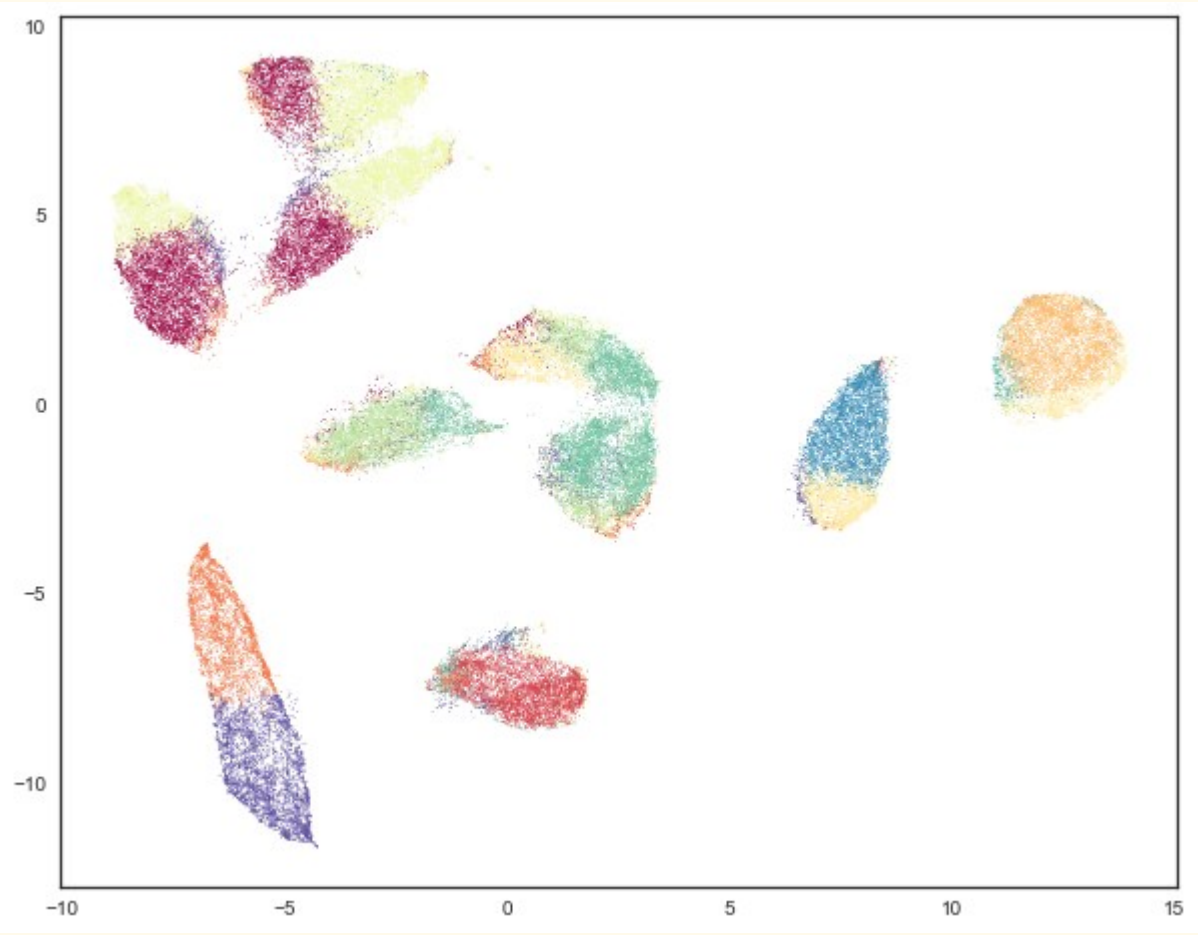
- Computationally efficient especially with big data
- It balances local and global structures
- Good at defining ecologically meaningful clusters
- Great python package

Caveats

- Can be unstable
- Inverse transform does not work very well
- Can sometimes produce artifacts
- Do not use with k-means
- R package has many limitations

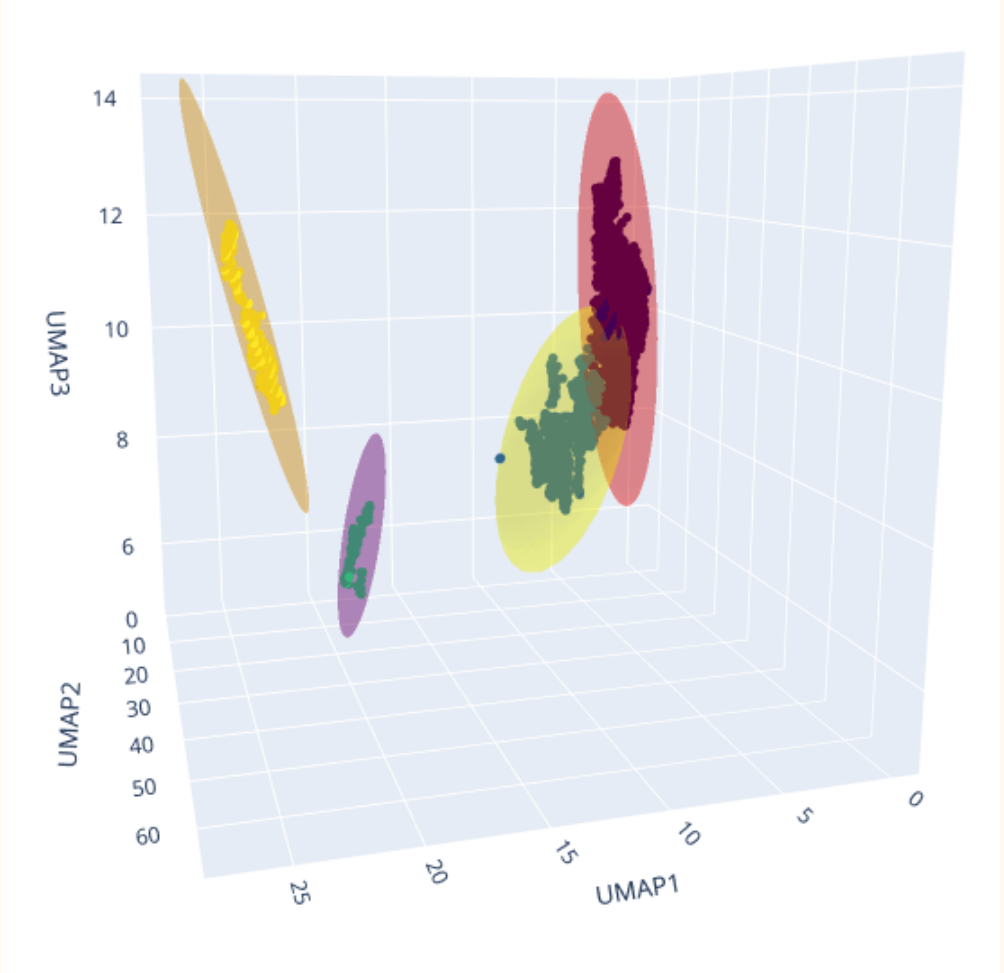
K-means with UMAP

- Does not work well with high dimensional space
- Do not recommend!



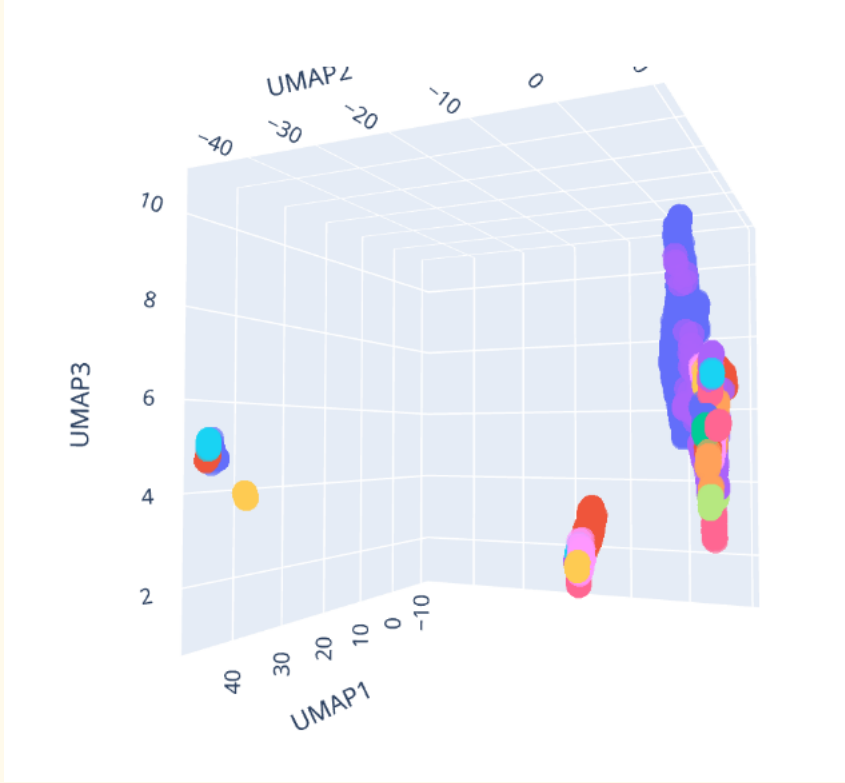
Hierarchical

- Works well with UMAP
  - (From my experience)
- Aligns well with ecology
- Caveats:
  - Not the best choice for complicated datasets
  - But can work with well separated clusters



Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN)

- Works but needs more samples to work with high dimensional data
- Gives a lot of clusters
- See website for example

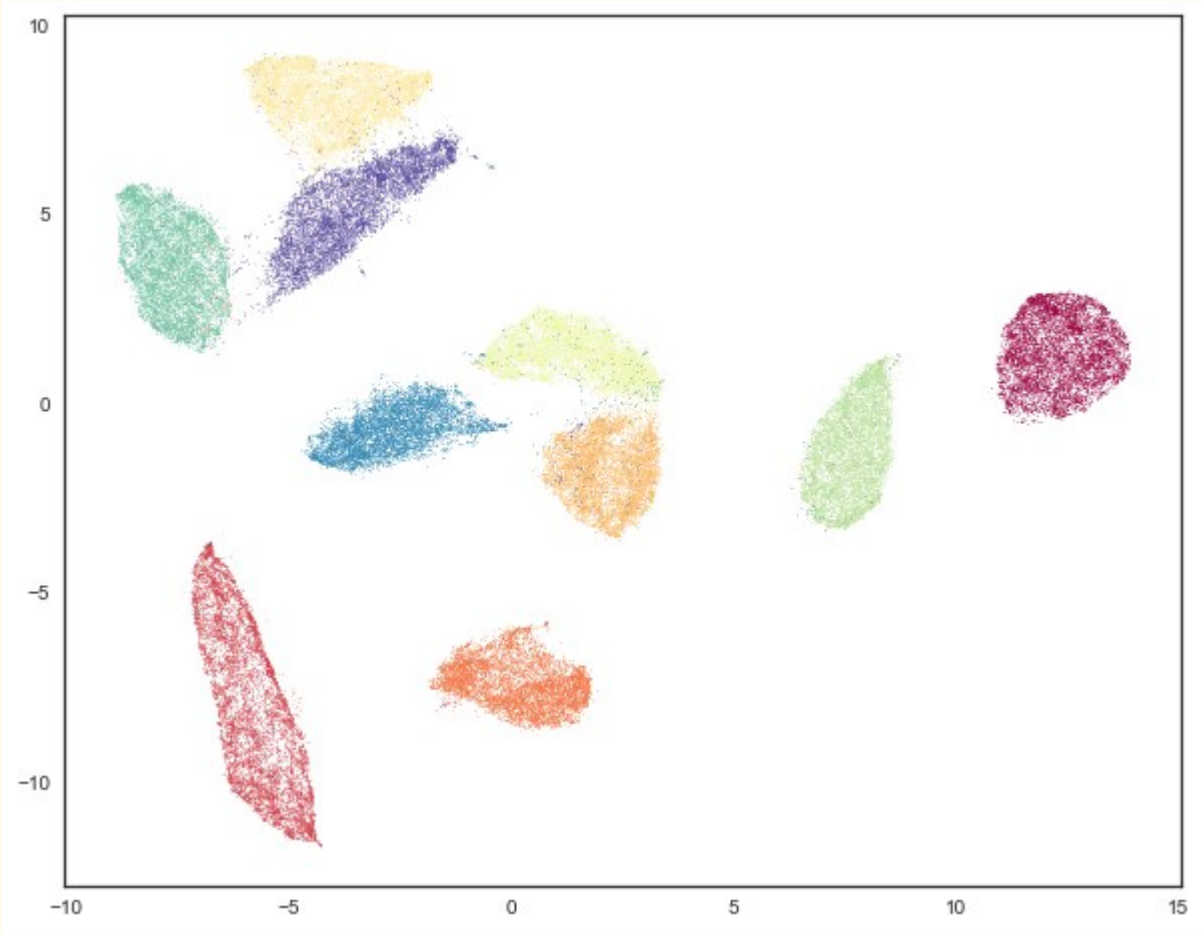


Other clustering methods

- Gaussian mixture models
- Self-organising maps (SOM)
- There are lots of options!

Creating clusters

- Should be attempted with care
- UMAP does not completely preserve density
- Can create false divisions



https://umap-learn.readthedocs.io/en/latest/clustering.html